

2026 FSAE: TECHNICAL INSPECTION

IMPORTANT: Each page must stay with the car until that specific part of inspection has been completed.

CAR NUMBER:

SCHOOL NAME: _____

Transponder number:

Present the vehicle for inspection in the following order:

- 1a. Supporting Equipment Inspection (bring all items from the IN.3 section in the next page)
- 1b. Mechanical Technical Inspection
2. Fuel Station & Tilt Table Inspection
3. Noise & Braking Performance Inspection



NOTE: If there is a conflict between this form and the rules, the rules prevail.

Inspector Names:

Non-Compliance Comments:

IC Electronic Throttle Special Inspection

ELECTRONIC THROTTLE CONTROL (ETC) - IC.4

•Only for cars using ETC: Do NOT signoff this tech form until separate ETC inspection is approved. •Requires prior submission and approval. Inspected by special team.

- Idle air bypass control and throttle blippers are ETC, and are not allowed without completing the ETC process.

Final Checklist:

- Structures / SES
- Templates
- Aero
- Mechanical
- Electronic Throttle

FINAL APPROVAL BY INSPECTOR:

DATE:

Supporting Equipment Inspection: IN.4

☐ 1. HELMETS - VE.3.2
•Closed-face with integral chin guard (no dirtbike helmets). •Face shield integral with helmet, impact resistant material. •Specification: Snell: K2015, K2020, M2015, M2020-D,-R, M2025-D,-R, SA2015, SA2020, SA2025; or SFI: 31.1/2015; 41.1/2015; or FIA: 8860-2010 & newer, 8859-2015 & newer. •No camera mounts: VE.2.5.3 •(Place sticker on left side)
☐ 2. BALACLAVA - VE.3.3.3
Required for all drivers. Flame Resistant material. Covers head, neck and hair. (No label required.)
☐ 3. ARM RESTRAINTS - VE.3.3.7
Required for all drivers. Must be commercially manufactured. (No label required.)
☐ 4. DRIVERS' SUITS - VE.3.3.1
Single piece suit, no holes. Must be labeled. Specification: SFI 3.2A/5; SFI 3.4/5; FIA 1986; FIA 8856-2000 or 8856-2018
☐ 5. GLOVES - VE.3.3.6
Flame Resistant material (not all-leather). No holes. Leather palms allowed only over flame resistant material. (No label required.)
☐ 6. SHOES - VE.3.3.5
No holes. Must be labeled. Specification: SFI 3.3; or FIA 8856-2000 or 8856-2018
☐ 7. SOCKS - VE.3.3.4
Flame resistant material (no cotton; no polyester). Must cover all bare skin. (No label required.)
☐ 8. WET TIRES - V.4.3.2
2.4mm min tread depth. Molded or cut by tire manufacturer. Brand, Compound: _____ Size: _____

Mechanical Inspection

☐ 9. FIRE EXTINGUISHERS - VE.2.3
•Hand-held, dry chemical (no AFFF or halon), min specification: 10BC; 1A10BC; 34B; 5A 34B; 20BE; or 1A 10BE. •Min 0.9 kg (2 lb). •Two required: must present both at Tech.
☐ 10. DRY TIRES - V.4.3.1
Brand, Compound: _____ Size: _____

Critical Fasteners Reference: T.8

Note: For all locations marked "CRITICAL FASTENERS" in this form, the following requirements apply: •Material rating: SAE Gr 5, Metric 8.8, or AN/MS. •Hex-head or socket-head; no pan head, button head, flat head, or low-profile screws. •Positive locking: safety wire, lock pins, or locking nuts with 2 threads visible beyond nut. Threadlock compounds not sufficient; no nylock nuts if >80 °C

•For any locking nuts used: must bring spares to Technical Inspection for examination (T8.3.3).

Mechanical Inspection: IN.6, Garage Entry

Reminder to teams: bring the following items: •Car on "Dry Tires" •Monocoque Laminate Test Specimens (if applicable) •The Impact Attenuator that was tested (if applicable) •Accumulator container and Tractive System firewall samples (if applicable) •Electronic, or printed, copies of any submitted Rules Questions including responses For egress (1 driver): •Helmet •Gloves •Arm Restraints •Long pants •Closed-toe shoes.
☐ 11. First Year Vehicle - GR.9.3
New Chassis and is within its initial Competition Year.
☐ 12. PUSH BAR - VE.2.2
Always with car (detachable), push & pull, usable by 2 people standing behind the car. EVs: HV disconnect tool, if used.
☐ 13. TRANSPONDER - VE.1.5
•Required type: AMB TranX 260; MyLaps X2, TR2. •Securely mounted with clear view of ground.
☐ 14. TRANSPONDER FUNCTION CHECK (if available)
☐ 15. Plastic RFID tag attached to left side of main hoop, facing leftward .

Driver In, Bodywork On

☐ 16. CAR NUMBERS - VE.1.1
On •front & both sides of car, •minimum 150 mm tall, •18 mm stroke & spacing. • 25 mm edge of number to edge of background •White-on-black or B-on-W only. •Background shape: round, oval, rectangular or square. •Visibility must not be obstructed.
☐ 17. SAE DECALS - VE.1.3
SAE logo on front OR both sides, in prominent location.
☐ 18. SCHOOL NAME - VE.1.2
School name or recognized initials. •Both sides of car, •easily visible location, •50 mm tall min, •Roman letters (or see VE.1.2.d), •high contrast background.
☐ 19. TECH STICKER SPACE - VE.1.4
25 cm wide x 20 cm high available space, located on centerline of upper front bodywork (nose) of car.
☐ 20. CAMERA MOUNTS - VE.2.5
•If >0.25 kg, must be secured by two points (typical GoPro-brand camera is <0.25 kg). •No cameras mounted to helmet. • No tube clamps on Main Hoop & Brace; requires weld tab at a node.
☐ 21. GROUND CLEARANCE - V.1.4
Sufficient clearance so that no part of the car other than the tires will contact the track surface.
☐ 22. BRAKE LIGHT - T.3.4
•RED color, •clearly visible from the rear, located on vehicle centerline. •Height between wheel centerline & driver's shoulders. •Round, triangle, or rectangular on black background. •15 cm ² minimum illuminated area. LED strips OK if elements closer than 20 mm apart and total length > 150 mm (5.9"). •Sufficient brightness for visible activation in bright sunlight.

Special Inspections (S#)

Structures Inspection: Instructions for teams:

- Bring the Registration Packet (this Tech Form).
- Participants must wear event wristbands.
- Remove bodywork and firewall prior to entry, sufficient for the inspection of impact attenuator, SIS, and V.1.4.2 ride height. Remove push bars and extinguishers during inspection.

Roll Hoops

- S1. MAIN HOOP - F.5.8 •Must be steel with smooth bends with no wrinkles. •Must be 1 piece & extend to lowest frame member. •Tube endpoints 380 mm apart (inside dim) at bottom attachment. •Above Upper SIS, max 10° to vertical. •Bottom ends, no more than 10° rearward from vertical.
- S2. MAIN HOOP BRACING - F.5.9, F.6.6 •Must be steel w/ no bends. •One brace each side, attached within 160 mm of top of hoop. •Min 30° included angle with hoop. •Bracing must not be on same side of vertical as Main Hoop.
- S3. HEAD RESTRAINT - T.2.8 •Inside Rollover Envelope or F.5.10 Head Restraint protection. •**Nothing can mount to the restraint support (no intake manifolds etc).** •Must take 900 N rearward and 300 N lateral forces **at the worst-case contact point.**
- S4. EXTERNAL ITEMS - F.5.11 (e.g. wings, suspension mounts, radiator, **cameras**) Single fastener connection at MH-MHB node preferred. For mid-MHB or mid-Upper-MH mounts: SES failure calculations and structural brace required. Light items may freely attach **internal** to tubes per F.5.11.3. Components offset longitudinally, not resting on tube. Suspension **and powertrain** mounts only permitted at nodes. **Clamshell mounts prohibited on MH and MHB.**
- S5. FRONT HOOP - F.5.7 •Closed-section metal tube (may be multi-piece). •Non-ferrous tubes must have inspection holes. •Must extend to lowest frame member. •Above Upper SIS, max 20° to vertical, F.6.3.5 max 10° towards cockpit if no rearward bracing.
- S6. **BENDS IN ROLL HOOPS – F5.6.3**
 - Front- and side-view bends *below* the respective Roll Hoop Braces must be: triangulated in side-view OR less than 25mm from a monocoque attachment point.

Chassis Features

- S7. STEERING COMPONENTS OUTSIDE FRAME - F.5.14
 - Rack inside primary structure. •F.5.14 Steering rack outside primary structure protected to top, bottom, and local chassis width. •Welded to chassis, or 30kN per end (60kN total) shown in SES.
- S8. LV BATTERIES, PRESSURIZED CYLINDERS - T.9.2.1, T.6.1.7
 - Inside F.1.1 chassis •Below Shoulder Harness. •Below top of chassis ahead & behind & to the left & to the right •F.1.13-15 envelopes outside chassis not legal.
- S9. HOLES AND OPENINGS IN REGULATED TUBES - F.5.3
 - Holes in regulated tubes addressed in SES.
 - Welded inserts should not fully interrupt tubes.**
 - Tube ends must be closed using welded caps or welded inserts, except bottom of roll hoops.

- S10. BOLTED JOINTS CONNECTING PRIMARY STRUCTURES F.5.4, F.5.13 •Double-shear (capped) or Sleeved butt joints only.
 - No bends. •No rod-end bearings. •Edge of any bolt hole located > 1.5 x hole diameter from nearest edge of the material. •No blind or welded threaded fasteners. •Bolts 8 mm (5/16"), plates 2.0 mm (0.08").
- S11. HARNESS ATTACHMENTS •T.2.4.3 Tube: 15kN, single shear per SES. •F.7.9 Composite: Test specimens, representative of vehicle. •Test loading direction: prefer normal, min 45 up from panel or harness installation per rules. •Tabs 1.6 mm (0.063") thick min, meets tear-out strength requirements in SES. Double-shear preferred. •**Cross tubes used for harness attachment must run straight across, between the Lower SIS.**

Front Impact

- S12. IA LEADING EDGE VERTICAL POSITION - F.8.5.6 •Bottom leading edge of IA must be lower than 175mm above lowest point of bottom surface of lower SIS tube. •Honeycomb must be horizontal; foam may be vertical. •V.1.4.2 Ride Height below lowest point of Lower SIS should be <=75mm, and must be <=90mm.
Record ride height measurement: _____
- S13. STANDARD IMPACT ATTENUATOR - F.8.4 •Must be fully bonded to AIP. •IA must not be degraded or damaged.
- S14. NON-STANDARD IMPACT ATTENUATOR - F.8.4.2
 - 200 mm long x 200 mm wide x 100 mm high. •Adhesive attachment required for foam or honeycomb. •Attached to AIP w/ welds - 1:1 weld ratio, beads 25 mm min, or eight 8 mm (5/16") bolts. •Must be capable of taking vertical and transverse loads. •Must be identical to test specimen, engraved dates of test. •Suitable failure mode (crushed element, not collapsed mount).
- S15. ANTI-INTRUSION PLATE - F.8.2
 - 1.5 mm steel or 4 mm aluminum. •120KN physical test, or 3-point composite per SES. •Attached to bulkhead w/ eight 8 mm (5/16") bolts, or bonded or laminated (AIP extending to outer edge of bulkhead), or >50% welded (AIP extending to centerline of bulkhead tubes). •Quick release and shear pin per SES if present. •CRITICAL FASTENERS: F.8.2.3.b
- S16. FRONT WING DETACHMENT - F.8.6 •No portion of front wing or mountings block the front view of bulkhead. •F.8.6.2 Detaches below peak load. •Fasteners match SES documentation.
- S17. FRONT BULKHEAD DIAGONAL - F.8.4.3
 - Unless tested for <25mm AIP deflection, diagonal structure is required for the standard foam IA when FBH > 400mm x 350mm; diagonal required for all standard honeycomb IAs.
 - Diagonal may be attached to **back** of AIP.

Tube Primary Structure

- S18. LOAD PATHS AND TRIANGULATION - F.1.17 •Upper and lower load paths, with sufficient diagonal tubing to fully triangulate in between, are required continuously through the FBHS, SIS, MHBs. •Minimum tube size requirements apply to Upper, Lower, and Diagonal structural tubes. •**MHB should be mitered to Lower MHBs.**

Tube Primary Structure (Continued)

- ☐ S19. BENT OR MULTIPLE TUBES - F.5.2
 •Min bend radius at centerline: 3x tube OD. •A brace must connect from on the bend to a frame node, •Within 30° of plane of bend (Upper SIS, Shoulder Harness Bar exempt from 30°).
 •Brace tube matches min, except Roll Hoop bends Size C, Upper SIS diag Size B, Shoulder Harness Brace Size C. •**No open tube-ends** in regulated load paths.
- ☐ S20. FRONT BULKHEAD SUPPORT - F.6.2
 •Connect bottoms of Front Bulkhead and Front Hoop; •Connect within 50mm of top of bulkhead, 100 mm above and 50 mm below upper SIS tube •**Rearward** brace to triangulated SIS if top tube does not connect near upper SIS.
- ☐ S21. FRONT HOOP BRACING - F.6.3
 •Two forward facing braces, attached within 50 mm of top of hoop, attached to Front Bulkhead or triangulated structure.
 •Rearward bracing if FH more than 10° rear above Upper SIS.
- ☐ S22. SIDE IMPACT STRUCTURE - F.6.4
 •Top surface of upper tube must be between 290 mm and 345 mm above bottom surface of lower SIS tube at its lowest point (F.6.4.4). •Lower tube must connect endpoints of Main and Front hoops.
- ☐ S23. SHOULDER HARNESS MOUNTING BAR - F.6.5
 •Uncut, closed-section tube. •Attached to Main Hoop.
 •Braced to Main Hoop if bent: minimum 30° between brace and bar. •Nothing attached except for Shoulder Harnesses, driver equipment, headrest, and light bodywork including the firewall and or seat.
- ☐ S24. TUBE MINIMUMS - F.3.2
 Tech may use ultrasound to measure wall thickness and/or ask that 4 mm dia holes be drilled.
 SIZE D ☐ Bent Upper SIS
 SIZE A ☐ MH ☐ FH ☐ SH
 SIZE B ☐ MHBrace ☐ Lap/Anti ☐ FB
☐ FHBrace ☐ FHBrace>10Rear ☐ SIS
 SIZE C ☐ IADiag ☐ FBHS ☐ MHBS ☐ SHBrace ☐ ExternalSteering

Monocoque

- ☐ S25. LAMINATE TEST SPECIMENS - F.4.3
 •Three-point bending: 275 x 500 mm or 138 x 500 mm.
 •Perimeter shear: 100 x 100 mm or 3-point. •Lap joint.
 •Engraved with date, specimen name.
- ☐ S26. BOLTED ROLL HOOP ATTACHMENTS - F.7.4, F.7.6
 •Three points per side x 30 kN each, or two x 45 kN. •Within 25mm of bends. •Boxed tube to plates and gussets •Mounting plates on hoop min 2mm thick. •Backing plates per SES, receipts for non-steel. •Two 8 mm bolts per joint. •One 10mm bolt on centerline allowed for hoop braces. •No blind or threaded inserts. •Requires inserts or no gap between inner and outer skins. •**MHBS tube nodes must be within 25mm of a roll hoop attachment.**
- ☐ S27. LAMINATED FRONT HOOP - F.7.4
 •Front Hoop may be fully laminated instead of bolting. •Must not be attached only by adhesive; cores must fit tightly. •**25mm inspection opening at bottom of front hoop lamination must be present for ultrasonic or drilled-hole tube wall inspection.**

- ☐ S28. SIDE IMPACT PROTECTION - F.7.5
 •Min 290mm height above bottom surface of chassis. •No side view openings between Front Hoop and Main Hoop.
- ☐ S29. LAMINATE THICKNESS VS SAMPLES AND SES - 4mm hole or external calipers.
 PANELS
 •FB •FBDiag •FBHS •FHBrace •SIS •Lap/Anti •SH •MHBS
 ATTACHMENTS
 •SteeringProtection •AIP •FH •Lap/Anti •SH •MH •MHB

Cockpit Templates

- ☐ S30. MAIN HOOP & FRONT HOOP HEIGHTS - F.5.6.5
 Helmet of 95th percentile male (PERCY) to be 50 mm below the lines between top of front and main roll hoops and between top of main hoop to rear attachment point of main hoop bracing. Center of bottom circle placed minimum 915 mm from pedals.
- ☐ S31. COCKPIT OPENING - T.1.1
 Template to pass from above cockpit to bottom of top SIS tube and less than 320mm from lowest point inside cockpit. Steering wheel & column, seat and padding can be removed; no removing firewall. Fore/aft translation of template OK.
- ☐ S32. COCKPIT INTERNAL CROSS SECTION - T.1.2
 Template to pass from rearwards of the steering column to 100 mm rearwards of the pedals.
 Steering wheel may be removed; padding may be removed if removable with no tools & with driver in seat.

Aero

- ☐ S33. AERODYNAMIC DEVICES - T.7.1
 •No powered ground effects. •Securely mounted - no oscillation or excessive movement. (Wings, undertray, splitter, endplates, vanes, etc.)
 •Vertical keepout zones 75mm in front & behind tires.
- ☐ S34. EDGE RADII - T.7.1.4-5
 •Horizontal leading edges min 5 mm radius.
 •Vertical forward-facing edges min 3 mm radius.
 •Other edges: not sharp - GR.1.5
- ☐ S35. FRONT MOUNTED AERO - T.7.5-8
 •Max forward: 700 mm ahead of the front tires.
 •Max width: OUTSIDE of the front tires (at hub height).
 •Max height: 250 mm if in front of tires (w/out driver).
- ☐ S36. REAR MOUNTED AERO - T.7.5-8
 •Max rearward: 250 mm behind the rear tires.
 •Max forward: Headrest support (undertrays exempt).
 •Max width: INSIDE of the rear tires (at hub height).
 •Max height: 1.2 m above ground (w/out driver).
- ☐ S37. AERO BETWEEN WHEEL CENTERLINES - T.7.5-8
 •Max width: Line between front & rear tires (at hub height).
 •Max height: 500 mm (exempt if w/in 400 mm of car centerline).

Inspector notes:

S12 IA LEADING EDGE VERTICAL POSITION: _____

S22 SIDE IMPACT STRUCTURE: _____

Time In: _____

Time Out: _____

Driver Fit

☐ 23. VISIBILITY - V.2.2 100° min field of view to each side. Head rotation OK, or mirrors. If mirrors, must be firmly installed and adjusted.
☐ 24. ROLL BAR PADDING - T.2.9 Installed on any bar that could be hit by the driver's helmet. •12 mm thick. •Specification: SFI 45.1 or FIA 8857/2001 (pipe insulation or other foams not OK).
☐ 25. OTHER SIDE TUBES - F.5.15 Cockpit design must prevent driver's neck from hitting bracing or other side tubes.
☐ 26. VEHICLE CONTROLS - T.1.4 No hands, arms, or elbows outside side impact system when actuating controls. All controls, including shifter, must be inside cockpit and below topmost point of Front Hoop.
☐ 27. LAP BELT FIT - T.2.5 •Must pass over pelvis, not waist. •45-65° to horizon for upright driver, 60-80° for reclined.
☐ 28. SHOULDER HARNESS FIT - T.2.6 Angle from shoulder between 10° up and 20° down (vs horiz).
☐ 29. SUB BELT FIT - T.2.7 •Snug, holding latch in place. •Position in side-view: 5 Point: aligned with or forward of shoulder belt line; 6 Point: vertical or rearward of latch.
☐ 30. ARM RESTRAINTS FIT - VE.3.3.7 Installed so the driver can release them and exit unassisted regardless of vehicle's position.
☐ 31. HEAD RESTRAINT FIT - T.2.8 •Max 25 mm (1") forward gap to helmet. •Helmet contact point min 50 mm (2") from any edge. APPLIES TO ALL DRIVERS (may be adjusted for each driver)
☐ 32. MAIN HOOP & FRONT HOOP HEIGHTS - F.5.6.4 Helmet 50 mm (2.0") below lines between •top of front and main roll hoops, and •top of main hoop to rear attachment point of main hoop bracing. (Applies for every driver.)
☐ 33. DRIVER'S FOOT PROTECTION - T.1.3.2 Feet must be rearward of the Front Bulkhead and no part of shoes or legs above or outside the Major Structure in side or front views when touching pedals. Remove nose bodywork if necessary for visual access.
☐ 34. EGRESS - IN.9.2 5 seconds max to actuate cockpit master switch and exit to side of vehicle, from driving position, wearing safety equipment. Wings must remain fixed in position. (See egress worksheet.)

Driver Out

☐ 35. BODY & STYLING - V.1 Open cockpit, formula style body. Four wheels, not in a line.
☐ 36. OPEN WHEEL - V.1.1 •Top 180° of wheel/tire unobstructed from above. •Tires unobstructed from sides.

☐ 37. BODYWORK - T.7.2 •No large openings (> ~6mm) in bodywork into driver compartment in front of or alongside driver. •Body/nose min 38 mm radius, +/-45° all directions.
☐ 38. WHEELBASE - V.1.2 Minimum 1525 mm.
☐ 39. JACKING POINT - VE.2.1 •Horizontal, lateral tube at the rear. •Orange color. •300 mm wide by 25-30 mm O.D. •Visible to person standing 1 meter behind car. •75 mm min ground clearance •Rear tires must come off the ground using Quick-Jack (lifts to 200 mm).
☐ 40. SEAT - T.1.5, T.1.6 •Insulated against heat conduction, convection and radiation. •Lowest point no lower than <u>top</u> of lower structure OR must have longitudinal 1.00" OD x 0.065" steel tube underneath.
☐ 41. FIREWALL - T.1.8 Rigid, nonflammable material. •Separates driver (line-of-sight to mid-height of driver's helmet) from fuel, cooling, oil, lithium battery, heat sources, and all tractive system components. Wire/cable pass-throughs OK with grommets. Multiple panels OK w/ gaps sealed. •Seals (e.g.: metal tape) cannot be stressed or used to hold firewalls in place. •No gaps at sides or bottom. •Leaks cannot contact driver.
☐ 42. FLOOR CLOSEOUT PANEL - T.1.7 Non-perforated, non-brittle material from foot area to firewall. Multiple panels OK if gaps less than 3 mm (1/8").

Bodywork Off

☐ 43. DRIVER'S LEG PROTECTION - T.1.3.3 Covers inside cockpit over sharp parts or moving suspension and steering components.
☐ 44. NON-CRUSHABLE OBJECTS - F.8.6.3 25 mm clearance aft of AI Plate, Front Bulkhead including diagonals, and to pedals @ full travel and adjustment.
☐ 45. THROTTLE PEDAL - IC.3.3.1 Must have positive stop to prevent overstressing cable. Must return when not actuated.
☐ 46. BRAKE PEDAL - T.3.2 Steel, aluminum, or machined titanium (no welded Ti). Alt matl. OK for pad face. Entire pedal-to-frame system capable of 2kN (tested only by organizers).
☐ 47. BRAKE PEDAL OVER TRAVEL SWITCH - T.3.3 •Must cut ignition & fuel pump. •No re-start if brake released or actuated a second time. •Must NOT rely on software to work. •Not resettable by driver•.
☐ 48. WHEELS - V.4 •203 mm (8.0") min diameter. •Wheels with single wheel nut must have positive retainer. •Aluminum lug nuts hard-anodized and pristine condition.

Driver Restraints

☐ 49. DRIVER RESTRAINT HARNESS - T.2.2 5, 6 or 7 point and be •labeled: SFI 16.1, 16.5, or FIA 8853/2016 (or lookup FIA D-number) •All lap belts must have Quick Adjusters. •Reclined drivers must have 6 or 7 point, and Quick Adjuster sub-belts OR 2 sets of sub belts. •Belts expire after expiration year (not month) marked on label (SFI & FIA).
☐ 50. HARNESS HARDWARE AND INSTALLATION - T.2.2 •Belts threaded through hardware per mfr instructions. •Hardware must be unmodified (no drilling, welding, etc).
☐ 51. HARNESS MOUNTS - T.2.4 •Belts must be protected by firewalls. •Tabs aligned with load direction of belt; may be pre-bent. (Mount structure requirements listed separately in “S11” above.) • Belts must be separated from the ground by the floor.
☐ 52. LAP BELT POSITION - T.2.5 •Pivoting mounting using eye bolt or shoulder bolt (no tube wrap). •Not re-directed by seat. •Belt centerline max 3” forward of seatback-seatbottom junction in side view.
☐ 53. SHOULDER HARNESS POSITION - T.2.6 Mounting points 175-235 mm (~7”- 9”) apart. Tube wrap OK.
☐ 54. SUB BELT POSITION - T.2.7 Sub belts cannot be re-directed by anything other than the driver's body.
☐ 55. BELT ATTACHMENT FASTENERS - T.2.5.8, T.2.6.4, T.2.7.5 •Lap & Shoulder 10 mm Grade 8.8 (3/8” SAE Grade 5), •Sub 8 mm (5/16”), or as specified by harness manufacturer. •Pins required in clip-brackets •Special rules for eye-type attachments: aligned, fully tight, see T.2.4.5 •CRITICAL (positive locking) - T.8
☐ 56. HEAD RESTRAINT - T.2.8 •Min 150 mm width •235 cm ² area AND height adjustment of 175 mm (7”); OR 280 mm (11”) height. •38 mm (1.5”) thick. •Near vertical. •Pad and mount must take 900 N rearward and 300 N lateral forces at the worst-case contact point . •Energy absorbing material: SFI Standard 45.2 or FIA Tech List 17 (CONFOR blue CF45 or CF45M), with 20mm inspection hole in its cover. •Mount must be within Main Hoop Bracing envelope, or add extra tube per F.5.10.

General Inspection - Chassis

☐ 57. STEERING - V.3.2 •All steerable wheels must have positive stops to prevent linkage lock-up or contact with other parts. •7° max freeplay at the steering wheel. •NO STEER-BY-WIRE on front wheels. •No cables or belts. •No bonded joints in column without metal backup. •Rear steer limited to 6° total, with mechanical stops.
☐ 58. CRITICAL FASTENERS, Steering - V.3.2.8 Steering wheel, column, rack mounting, tie rods.
☐ 59. SUSPENSION PICK-UP POINTS - V.3.1.3, GR.1.5 Inspected thoroughly for integrity: binding, over-articulation.

☐ 60. STEERING WHEEL - V.3.3 •Continuous perimeter, near round (no concave sections). •Driver operable quick disconnect. •Not higher than top of Front Hoop, in any angular position. •250 mm max rearward of Front Hoop (F.5.7.5).
☐ 61. SUSPENSION - V.3.1 •Full suspension including front and rear damping. •Spherical rod ends and bearings: double-shear or safety washers.
☐ 62. MODIFIED LUG BOLTS/STUDS - V.4.2.2 Verify good engineering practices are followed e.g. no drill holes for mass reduction.
☐ 63. CRITICAL FASTENERS: SUSPENSION - V.3.1.4 Control arms, knuckle, spring load path, single wheel nuts. Exempt: lug nuts, multi-piece wheels, anti-roll bars, dampers. All fasteners must be tight (esp: jam nuts).
☐ 64. BRAKES - T.3.1 •Single pedal actuates all 4 wheels (one brake on limited slip OK) •Two separate hydraulic circuits w/ reservoirs; no brake-by-wire. •Protected by structure/shields from drivetrain & collisions. •No plastic brake lines. •No parts below chassis/tub in side-view.
☐ 65. CRITICAL FASTENERS: BRAKES - T.3.1 •Pedal Assembly: including adjustment mechanism, caliper to knuckle mounts, rotor to hat. No nyloc nuts near disc or caliper. •Exempt: COTS caliper body assembly
☐ 66. BRAKE CONTROL SYSTEMS - GR.1.5 ABS, Traction, Yaw Control, etc: must be approved by a Tech Chief.

IC Powertrain

☐ 67. ENGINE - IC.1.1 Four-stroke piston engine, 710 cc maximum swept displacement. No hybrids. Waste heat recovery is allowed.
☐ 68. COOLANT - T.5.4 Only 100% water. NO ADDITIVES WHATSOEVER.
☐ 69. INTAKE and FUEL SYSTEM LOCATION - IC.1.2 •All parts of air intake system (including throttle body or carb, air intake ducting, air cleaner & air box), AND •all parts of the fuel storage, supply and fuel control systems (including fuel rail, throttle body or carburetor), must be within a surface defined by the top of the roll bar and the outside top edge of the tires.
☐ 70. AIR INTAKE SYSTEM - IC.2.2 & .3 •Side and Rear Impact protection if <350 mm above ground. •Supported if cantilevered (isolated to frame, rigid to engine). •CRITICAL FASTENERS: securely attached to block or head with brackets & mechanical fasteners w/ positive locking mechanisms. OEM-type rubber bushings not sufficient.
☐ 71. THROTTLE - IC.3 •Min qty of 2 springs at the TB, each capable of closing the throttle independently. TPS not acceptable as a return spring. •Cable must have smooth operation with no binding or sticking. •Cable position min 50 mm from any exhaust component. •Idle air bypass control and throttle blippers are ETC and not allowed without ETC process.

IC Powertrain (Continued)

☐ 72. RESTRICTOR - IC.2.4
•Must be circular: max dia 20.0 mm for gasoline and 19.0 mm for E85. •Cannot be movable or flexible. •Installed per below: NA: THROTTLE -> RESTRICTOR -> ENGINE FI: RESTRICTOR -> COMPRESSOR -> THROTTLE -> ENGINE
☐ 73. HIGH PRESSURE HYDRAULICS - T.6.2
Pumps and lines must have 1 mm thick steel or aluminum shields to protect driver and workers. (Brakes & clutch exempt.)
☐ 74. COMPRESSORS - IC.2.5
•Turbo or super chargers allowed if not OEM to engine. •Must be between restrictor and throttle. •Intercoolers downstream of throttle. •Carbs not allowed if compressors are used. •Compressor recirculation valves ok if downstream of restrictor. •No enlarged air chambers (section > 28 cm²) before throttle.
☐ 75. CATCH CANS - T.5.6
•Engine coolant (unless aircooled) and engine crankcase must have separate catch cans of 0.9 L min vol. •Oil(s) and water(s) must be separate. •100 °C-capable material. •Behind firewall, below shoulder level. •3 mm min diameter vent, directed away from driver. •PCV OK if routed to intake sys upstream of restrictor. •Cannot connect breathers to exhaust. •Trans, diff, other systems (unless sealed): 10% or 0.5 L catch can.
☐ 76. FLUID ACCUMULATION - T.5.5.5
Absorbent materials and open collection devices (regardless of material) are prohibited below the highest point of the exhaust system in compartments containing the engine, drivetrain, exhaust and fuel systems.
☐ 77. BELLYPANS - T.5.5.4
Must be vented to prevent accumulation of fuel: 2 holes each min of 25mm dia. •Specific locations: Lowest point in chassis; Aft of driver & front of fuel tank.
☐ 78. FLUID LEAKS - T.5.5.1
Not permitted. Firewalls must prevent contact w/ driver.
☐ 79. EXHAUST OUTLET - IC.7.2
•Outlet 45 cm (17.7") max behind rear axle centerline and 60 cm (23.6") max above the ground. •Located such that exhaust gases should not reach driver.
☐ 80. EXHAUST SYSTEM - IC.7.2
•Exhaust components outside bodywork forward of main hoop must be shielded from people approaching the car. •No absorbent wraps around exhaust tubes.
☐ 81. SCATTERSHIELDS GENERAL - T.5.2
•Required for clutches, chains, belts, CVT rotating parts, etc. •Not perforated. •End parallel to lowest part of front and rear sprockets. •Must protect fuel, brake, and pneumatic components. •Min 6mm fasteners •CRITICAL FASTENERS
☐ 82. SCATTERSHIELD MATERIALS - T.5.2
•Size: for chains: 2.7 mm (0.105") min thick steel, 3x chain width; for belts: 3 mm (0.12") min thick aluminum 6061-T6, 1.7x belt width. •OEM engine drive sprocket cover OK.

☐ 83. D'TRAIN FINGER GUARDS - T.5.2.10
Required to cover all drivetrain parts that spin while car is at rest. No holes >12 mm dia.
☐ 84. COMPRESSED GAS CYLINDERS - T.6
•Unmodified COTS cylinder (labeled). •Nonflammable gas. •Regulator on tank. •Appropriate lines & fittings. •Securely mounted, axis not pointed at driver. •Rearward of firewall within the frame envelope, or in structural sidepod; not in cockpit. •Protected by T.1.6.3 when <=150mm from any hot parts.

Fuel System

☐ 85. FUEL TANKS - IC.5.2 & .3, F.9
•Must lie within major structure of the chassis, with side impact protection. •Rigid tanks cannot carry structural load & must be flexibly mounted. •Bladders or bags in rigid container. •No portion of fuel system below lower surface of frame. •Firewall between all parts of fuel system & driver.
☐ 86. FUEL LINES - IC.5.7
•No plastic lines between tank & engine (reinforced rubber hoses OK). •Bulbs/barbs on hose connections. •No worm-gear clamps. •Must be securely attached, •Protected from rotating equipment & collision damage. •Systems >10 bar see IC.6.2
☐ 87. GOOD PRACTICES, fuel lines - GR.1.5
•Hoses and fittings must be type-matched (no clamps on braided metal hoses, etc). •Fuel lines restrained and protected from stress, heat, and abrasion.
☐ 88. FUEL FILLER NECK - IC.5.4
•Fuel-resistant materials, •min 35 mm inner dia, •within 30° of vertical. •Must prevent fuel spillage contacting driver, exhaust or ignition (add shields as needed). •Fueled w/o manipulating car in any way. •Cap secure and capable of withstanding pressurization (ie: threads or latch). •Easy access for common 2-gal jugs.
☐ 89. SIGHT TUBE - IC.5.4
•Fuel resistant materials, •transparent, •min 6mm inner dia. •Min 125 mm vertical height in area visible to fueler with vehicle fully assembled. •Sight tube must NOT run below top of tank. •Non-moveable fuel level line 12-25 mm below top of sight tube. (Clear filler neck OK as sight tube.)
☐ 90. FUEL RAIL - IC.6.1.2 (LP), IC.6.2.4 (HP)
•Securely attached to block, head or intake manifold with brackets & mechanical fasteners. •No plastic or composite fuel rails, except if unmodified OEM part. •CRITICAL FASTENERS
☐ 91. FUEL TYPE - IC.5.1
•Circle type: 93 octane gasoline 100 octane gasoline E-85 - No agents other than the provided fuel and air may be induced into the combustion chamber. •Place appropriate fuel sticker adjacent to fuel filler (not on cap).
☐ 92. FUEL VENTS - IC.5.6
•Must exit outside of the bodywork. •Must include a check valve to prevent leakage if car inverted.

Electrical

☐ 93. ON-BOARD STARTER - IC.8.1 Required (remote starters and push-starts prohibited).
☐ 94. PRIMARY MASTER SWITCH - IC.9.3 •On driver's right, near roll bar. •Access from outside of car. •Rotary type. •No relay. •Must kill ALL electrical systems. •Marked with international symbol. •Lever horizontal when ON.
☐ 95. COCKPIT MASTER SWITCH - IC.9.4 •Pull-ON, Push-OFF type. •Alongside & unobstructed by steering wheel, easily reached by driver. •Must kill ignition & fuel pump(s). •Min dia 24 mm. •Marked with international symbol.
☐ 96. BATTERY - T.9.2 •Attached securely inside the frame or chassis. •Hot terminal insulated. •Mounted lower than the shoulder harness mount. •Wet-cells in marine box if inside cockpit. •Type must be identifiable. •Overcurrent protection •Lithium: firewall per T.1.8 between driver; OEM battery or with rigid nonflammable case. •No circuits > 60 VDC.

Overall

☐ 97. GOOD ENGINEERING PRACTICES - GR.1.5 •Proper use of fasteners. •Proper use of fluid lines and fittings. •Appropriate selection of materials regarding fluids, heat. •Protection from sharp edges - wiring, hoses, people. •Protection from heat - wiring, hoses, people. •Linkages not bound up or prone to over-articulation. •No excessive lash in joints and pivots.
☐ 98. VISIBLE ACCESS - IN.2.3 To all items on Tech Sheet without the use of mirrors, borescopes, etc.

FORMULA SAE - DRIVER COCKPIT CHECKS

Car #	School Name:								
Driver's Name	Helmet Lines	Headrest: Fore-Aft	Headrest: To Edges	Lap Belt Position	Shoulder Belt Position	Sub Belt Position	Egress Time	Driver's License	Inspector Initials

Helmet Lines: 50 mm (2") min. below lines between Main & Front Hoops and between Main Hoop & rear attachment point of Main Hoop Bracing
Head Restraint: Fore & aft: 25.4 mm (1 in) max gap to back of helmet.
Head Restraint: To edges: Helmet contact point min 50 mm from any edge.
Lap Belt: Over hip bones and tight. Arm restraints connected to latch.
Shoulder Belts: Tight. 10° up thru 20° down (from shoulder, relative to horizontal).
If shoulder belts are labeled "FHR required" then driver must wear a neck support (e.g.: HANS).
Sub Belts: Tight. Side-view position: 5-point: aligned with or forward of shoulder belt line, 6-point: vertical or rearward of latch.
Egress: Max 5.0 sec from "go" to BOTH feet on ground. Must include actuation of cockpit master switch.
RFID Wristband: Before Tech, drivers must login and complete the My FSAE DriverPass user info at fsaeonline.com

2026 FSAE INSPECTION SHEET

CAR NUMBER:
SCHOOL:
ENGINE MODEL:
ENGINE BORE X STROKE:
ABS? YES/NO

IMPORTANT

THIS FORM MUST STAY WITH THE CAR UNTIL THESE PARTS OF INSPECTION HAVE BEEN COMPLETED

PART 2

FUEL SYSTEM & TILT TABLE INSPECTION

FUEL STICKER - Fuel sticker in place adjacent to F/T filler. Mark type of fuel used on this form. (e.g. 93, 100 or E-85)	FUEL INTEGRITY -System can be filled with no leaks observed, using standard fuel station equipment without special tools or assistance.
SPILLAGE - No fluid leaks of any kind permitted when car is tilted to 45 degrees in the direction most likely to create spillage; Fuel tanks must be filled to their sight tube fill line.	VEHICLE STABILITY - All wheels in contact with tilt table when tilted to 60 degrees to the horizontal.

NON-COMPLIANCE / COMMENTS:

APPROVED BY:

DATE:

PART 3

NOISE LEVEL & BRAKING PERFORMANCE INSPECTION

NOISE LEVEL - 110 dB (C) ("C" scale) maximum during a static test, gearbox in neutral, UP TO a specified RPM (see Rule IN.10.4.1). 103 dBC at idle. Microphone level with the exhaust outlet(s), 0.5 m (19.7") from the outlet(s), at 45 degrees to the outlet. If multiple outlets, all to be checked. If movable tuning or throttling device, see IN.10.2.3.	BRAKING PERFORMANCE - Must lock up all 4 wheels, come to a complete stop with them locked and keep the engine running (IC only). Electric vehicles must turn off ABS system before performing test to allow lockup.
MASTER SWITCH - Master switch on RHS of main roll hoop must cause engine to stop when actuated. (Perform at end of noise test)	
NOISE LEVEL:	ATTEMPTS:

NON-COMPLIANCE / COMMENTS:

APPROVED BY:

DATE: